

Acceptance Tests – WBOCRM

This document maps READ acceptance criteria to implemented test coverage.

All tests are located in backend/tests/ and results saved in backend/test_results/.

Test Suite Summary

Suite	File	Tests	Status
Unit	tests/unit/authService.test.js	6	■ PASS
Unit	tests/unit/scoringService.test.js	6	■ PASS
Unit	tests/unit/ticketService.test.js	6	■ PASS
Unit	tests/unit/leadService.test.js	12	■ PASS
Unit	tests/unit/dashboardService.test.js	6	■ PASS
Functional	tests/functional/api.test.js	15	■ PASS
BDD	tests/features/auth_login.feature	4 scenarios	■ PASS
BDD	tests/features/lead_registration.feature	2 scenarios	■ PASS
BDD	tests/features/lead_scoring.feature	3 scenarios	■ PASS
BDD	tests/features/ticket_creation.feature	2 scenarios	■ PASS
BDD	tests/features/duplicate_email.feature	2 scenarios	■ PASS
BDD	tests/features/interaction_logging.feature	2 scenarios	■ PASS
BDD	tests/features/pipeline_stage.feature	2 scenarios	■ PASS
BDD	tests/features/dashboard_kpis.feature	2 scenarios	■ PASS
BDD	tests/features/csv_export.feature	1 scenario	■ PASS
Total		53 tests / 20 BDD scenarios	■ All pass

BDD Acceptance Scenarios

Feature: Lead Registration (lead_registration.feature)

Scenario 1: Successfully register a new lead

Given I am authenticated as a sales user
When I submit a new lead with email "bdd_lead@test.wbocrm" and name "BDD Lead"
Then the response status should be 201
And the response should contain a lead_id

Covers: US-03, FR-LM-01

Scenario 2: Reject duplicate email registration

Given a lead with email "bdd_dup@test.wbocrm" already exists
When I submit a new lead with email "bdd_dup@test.wbocrm" and name "Dup Lead"
Then the response status should be 400
And the response error should be "DUPLICATE_EMAIL"

Covers: US-03, FR-LM-02

Feature: Support Ticket Creation (ticket_creation.feature)

Scenario 1: Successfully create a support ticket

Given I am authenticated as a support user
And a lead exists in the system
When I create a ticket with description "BDD test ticket" and priority "Low"
Then the response status should be 201
And the response should contain a ticket_id

Covers: US-11, FR-ST-01

Scenario 2: Update a ticket status

Given an existing support ticket
When I update the ticket status to "In Progress"
Then the response status should be 200
And the ticket status should be "In Progress"

Covers: US-12, FR-ST-02

Feature: User Authentication (auth_login.feature)

Scenario 1: Successful login with valid credentials

When I log in with email "bdd_auth_sales@test.wbocrm" and password "BddAuth123!"
Then the response status should be 200
And the response should contain a token
And the token payload should include role "sales"

Covers: US-01, FR-AUTH-01

Scenario 2: Failed login with invalid password

When I log in with email "bdd_auth_sales@test.wbocrm" and password "WrongPass999!"
Then the response status should be 401

Covers: US-01, FR-AUTH-02

Scenario 3: Rejected request without token

When I request the leads list without a token
Then the response status should be 401

Covers: US-01, FR-AUTH-03

Scenario 4: Rejected request with insufficient role

Given I am authenticated as a support user
When I request the leads list
Then the response status should be 403

Covers: US-02, FR-AUTH-04

Feature: Lead Priority Scoring (lead_scoring.feature)

Scenario 1: High engagement metrics receive high score

When I create a lead with metrics calls 20 meetings 5 budget "high" companySize "enterprise" emailOpens 20
Then the response status should be 201
And the lead priority_score should be greater than 70

Covers: US-04, FR-SC-01

Scenario 2: No engagement receives low score

When I create a lead with metrics calls 0 meetings 0 budget "low" companySize "small" emailOpens 0
 Then the response status should be 201
 And the lead priority_score should be less than 30

Covers: US-04, FR-SC-01

Scenario 3: Leads list sorted by priority score descending

When I retrieve all leads
 Then the response status should be 200
 And leads should be ordered by priority_score descending

Covers: US-04, FR-LM-03, FR-SC-01

Functional Test Coverage (`api.test.js`)

Test	Expected	Covers
POST /api/leads – valid lead	201 + lead_id	US-03
POST /api/leads – duplicate email	400 DUPLICATE_EMAIL	US-03
GET /api/leads – sorted DESC	200 + ordered array	US-04
POST /api/leads/:id/logs – add note	201 + note_text	US-09, FR-ST-04
GET /api/leads/:id/logs – get logs	200 + array	US-09, FR-ST-04
PUT /api/leads/:id – pipeline stage	200 + stage updated	US-05, FR-UC-05
GET /api/tickets – ticket list	200 + array	FR-ST-09
POST + PUT /api/tickets/:id	201 then 200 + status updated	US-11, US-12
GET /api/dashboard – KPIs	200 + all 4 fields	US-13, FR-ST-11/12/14
POST /api/leads – campaign_id	201 + campaign_id stored	FR-SC-13
GET /api/leads/export/csv	200 + text/csv header	US-07, FR-DOC-15
GET /api/leads/:id – PII masked (support)	200 + masked email/name	NFR-ST-07
GET /api/leads/:id – unmasked (sales)	200 + full data	NFR-ST-07
GET /api/leads – no token	401	US-01
GET /api/leads – wrong role	403	US-02

Unit Test Coverage

authService (6 tests)

- hashPassword returns a bcrypt hash
- comparePasswords returns true for matching password
- comparePasswords returns false for wrong password
- generateToken returns a signed JWT
- generateToken payload contains user_id and rbac_role
- verifyToken returns the original payload

scoringService (6 tests)

- Returns 0 for empty metrics
- Caps call score at 25 pts
- Caps meeting score at 25 pts
- Applies correct company size scores (small/medium/enterprise)
- Returns maximum 100 for fully maxed metrics

- Rounds to integer

ticketService (6 tests)

- Creates a ticket successfully
- Returns TICKET_NOT_FOUND for missing ticket
- Returns CONFLICT when updated_at does not match
- Updates status successfully on valid updated_at
- Validates status enum values
- Assigns ticket to user correctly

leadService (12 tests)

- Accepts valid pipeline stage "Contacted" (FR-UC-05)
- Throws INVALID_STAGE for unknown stage (FR-UC-05)
- Accepts all four valid stages: New, Contacted, Qualified, Closed (FR-UC-05)
- Updates user_id for lead assignment (FR-SC-06)
- Throws LEAD_NOT_FOUND when lead does not exist (FR-SC-06)
- Creates an interaction note via addLog (FR-ST-04)
- Throws LEAD_NOT_FOUND for addLog on missing lead (FR-ST-04)
- Returns interaction logs via getLogs (FR-ST-04)
- Stores campaign_id when creating a lead (FR-SC-13)
- Anonymises lead PII via erasePersonalData (NFR-ST-07)
- Throws LEAD_NOT_FOUND for erasePersonalData on missing lead (NFR-ST-07)
- Validates engagement metrics (INVALID_METRICS)

dashboardController (6 tests)

- Returns activeLeads count (FR-ST-11)
- Returns openTickets count (FR-UC-12)
- Returns monthlyRevenue sum (FR-ST-14)
- Returns all four KPI fields (FR-ST-11/12/14)
- Masks top5 PII for support role (NFR-ST-07)
- Does NOT mask top5 PII for admin role (NFR-ST-07)

Non-Functional Requirements Verified

NFR ID	Requirement	Verification
NFR-ST-05	Optimistic locking on ticket updates	Unit test: CONFLICT on stale updated_at
NFR-ST-12	Dashboard auto-refresh every 2s	DashboardPage setInterval(2000) implementation
NFR-SEC-01	Passwords hashed with bcrypt	Unit test: hashPassword returns non-plaintext
NFR-SEC-02	JWT auth on all protected routes	Functional test: 401 without token
NFR-SEC-03	Role enforcement	Functional test: 403 for insufficient role
NFR-ST-07	GDPR/KVKK PII masking	leadController masks email/name for support role
NFR-ST-07	Right-to-erasure	DELETE /leads/:id/personal-data, /users/:id/personal-data
NFR-ST-15	Ticket archiving (365+ days)	archiveService daily cron + manual POST /tickets/archive
NFR-SEC-04	Idle session timeout	AuthContext 30-min inactivity auto-logout

Feature: Interaction Logging (`interaction_logging.feature`)

Scenario 1: Successfully add an interaction note to a lead

```
Given I am authenticated as a sales user
And a lead exists in the system
When I add an interaction note "Initial phone call - very interested" to the lead
Then the response status should be 201
And the response should contain the note text "Initial phone call - very interested"
```

Covers: US-09, FR-ST-04

Scenario 2: Retrieve interaction logs for a lead

```
Given I am authenticated as a sales user
And a lead exists in the system
And I add an interaction note "Follow-up email sent" to the lead
When I retrieve the interaction logs for the lead
Then the response status should be 200
And the logs should contain at least 1 entry
```

Covers: US-09, FR-ST-04

Feature: Pipeline Stage Management (`pipeline_stage.feature`)

Scenario 1: Successfully update pipeline stage to Qualified

```
Given I am authenticated as a sales user
And a lead exists in the system
When I update the lead pipeline stage to "Qualified"
Then the response status should be 200
And the lead pipeline_stage should be "Qualified"
```

Covers: US-05, FR-UC-05

Scenario 2: Reject invalid pipeline stage

```
Given I am authenticated as a sales user
And a lead exists in the system
When I update the lead pipeline stage to "InvalidStage"
Then the response status should be 400
And the response error should be "INVALID_STAGE"
```

Covers: US-05, FR-UC-05

Feature: Dashboard KPIs (`dashboard_kpis.feature`)

Scenario 1: Dashboard returns all KPI fields

```
Given I am authenticated as a sales user
When I request the dashboard
Then the response status should be 200
And the dashboard should contain "activeLeads"
And the dashboard should contain "openTickets"
And the dashboard should contain "top5"
And the dashboard should contain "monthlyRevenue"
```

Covers: US-13, FR-ST-11, FR-UC-12, FR-ST-14

Scenario 2: Dashboard top5 contains leads sorted by score

```
Given I am authenticated as a sales user
When I request the dashboard
Then the response status should be 200
And the dashboard top5 should be an array
```

Covers: US-13, FR-ST-11

Feature: CSV Export (csv_export.feature)

Scenario: Successfully export leads as CSV

Given I am authenticated as a sales user
When I request the leads CSV export
Then the response status should be 200
And the response content-type should be "text/csv"
And the CSV should contain the header "lead_id,contact_name,email"

Covers: US-07, FR-DOC-15

Feature: Duplicate Email Detection (duplicate_email.feature)

Scenario 1: Reject lead with existing email

Given I am authenticated as a sales user
And a lead with email "existing@test.wbocrm" already exists
When I submit a new lead with email "existing@test.wbocrm"
Then the response status should be 400
And the response error should be "DUPLICATE_EMAIL"

Covers: US-03, FR-LM-02

Scenario 2: Frontend redirects to existing lead profile

Given I am on the leads page
When I create a lead with a duplicate email
Then I should see an error message containing "DUPLICATE_EMAIL"
And a link to the existing lead profile should be displayed

Covers: US-03, FR-LM-02

Additional Coverage Notes

The following features are verified via code review and manual testing (frontend-only or complex setup):

Feature	Implementation	Verification Method
Ticket archiving	archiveService.js	Code review + manual API test
Idle session timeout	AuthContext.jsx (30 min)	Code review + manual UI test
Offline draft caching	TicketPage.jsx localStorage	Code review + manual UI test
Last admin guard	userController.updateRole	Code review + manual API test